Description:

INPUT: take the source file

The Log Analyzer agent processes raw logs and converts them into structured insights by following a defined algorithmic workflow.

BEGIN

1. INPUT_HANDLING:
   IF input is plain text OR JSON OR .log OR .txt OR .csv OR .zip THEN
      Load input
   ELSE IF input is streaming chunks THEN
      Append chunks to buffer
   ENDIF
   Normalize line breaks and encoding.

2. PARSE_AND_NORMALIZE:
   FOR each log entry IN input:
      Detect format:
         IF JSON !' parse JSON fields
         ELSE IF Apache/Nginx !' extract IP, method, URL, status
         ELSE IF stack trace !' group multiline traces
         ELSE !' treat as generic log line
      ENDIF
      Extract timestamp, log level, source, message.
      Store normalized entry.
   ENDFOR

3. ERROR_WARNING_EXTRACTION:
   FOR each normalized entry:
      IF entry.level == ERROR OR WARN THEN
         Add to error_list
      ENDIF
      IF message contains keywords (Exception, Timeout, Failed, Invalid, Unauthorized, StackTrace) THEN
         Tag as critical_event
      ENDIF
   ENDFOR

4. ANOMALY_DETECTION:
   IF error frequency spikes above threshold THEN
      Mark anomaly("Error Spike")
   ENDIF
   IF repeated messages occur in short interval THEN
      Mark anomaly("Repeated Failures")
   ENDIF

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  IF resource-related patterns detected (Memory, CPU, Throttle) THEN
    Mark anomaly("Resource Issue")
  ENDIF

5. ROOT_CAUSE_ANALYSIS:
  FOR each critical_event:
    IF "Caused by" is present THEN
      Link event to cause
    ELSE IF event closely precedes other failures THEN
      Infer cause-effect chain
    ENDIF
  ENDFOR

6. PERFORMANCE_ANALYSIS:
  IF response_time > threshold THEN
    Add to slow_requests
  ENDIF
  IF timestamps show delay patterns THEN
    Flag latency issue
  ENDIF
  IF endpoints repeatedly slow THEN
    Mark bottleneck
  ENDIF

7. SUMMARIZE_AND_REPORT:
  Generate:
    Summary = overall system issue in 1–2 lines
    Key_Events = all high-impact log entries
    Root_Cause = most likely triggering factor
    Anomalies = unusual or repeated patterns
    Performance_Findings = latency, bottlenecks
    Security_Concerns = authentication/authorization failures
    Recommendations = clear troubleshooting steps
    Severity = (Critical / High / Medium / Low)
    Timeline = ordered events by timestamp

RETURN all structured fields.

END
```

You have access to the KnowledgeRAGTool. It is advised to use this tool when you have access to it irrespective of whether a knowledge retrieval task has been assigned explicitly.

Expected Output:

A comprehensive log analysis report with summaries, key findings, root cause indicators, and actionable recommendations.

Summary:

INPUT: take the source file

The Log Analyzer agent processes raw...

Raw Output:

Comprehensive Log Analysis Report

Summary:

Multiple critical errors observed across authentication, database connectivity, and service reliability. Frequent timeouts, connection resets, and authentication failures indicate systemic instability and security concerns.

Key Events:

- 2025-02-11 10:31:12 ERROR Timeout while querying database (30000ms)
- 2025-02-11 10:31:12 ERROR Caused by: DBConnectionError: db.example.net unreachable
- 2025-02-11 10:31:13 WARNING Slow request: /api/user took 1892ms
- 2025-02-11 10:31:14 ERROR Auth failure for user: token expired
- 2025-02-11 10:31:16 ERROR Timeout while querying database (30000ms)
- 2025-02-10T10:32:11Z ERROR Timeout after 30000ms (UserService)
- 2025-02-10T10:32:11Z ERROR Database connection failed
- 2025-02-10T10:32:12Z WARN Retrying request (attempt 1)
- 2025-02-10T10:32:15Z ERROR Invalid token for userID 2421 (AuthService)
- 2025-02-10T10:32:18Z INFO Request processed in 8290ms
- 2025-02-11T14:10:07Z ERROR OrderService failed: ECONNRESET
- 2025-02-11T14:10:08Z ERROR OrderService failed: ECONNRESET
- 2025-02-11T14:10:10Z WARN Processing delay: 5200ms
- 2025-02-11T14:10:14Z INFO Request completed with latency 6300ms
- 2025-02-09T09:12:01Z WARN Multiple login attempts for user admin
- 2025-02-09T09:12:02Z ERROR Authentication failed for user admin
- 2025-02-09T09:12:04Z ERROR Authentication failed for user admin
- 2025-02-09T09:12:05Z INFO IP blocked due to repeated invalid attempts: 192.168.1.34
- 2025-01-28T07:45:11Z ERROR Unhandled exception occurred (PaymentService)
- 2025-01-28T07:45:11Z ERROR java.lang.NullPointerException at PaymentProcessor.java:42
- 2025-01-28T07:45:12Z INFO Service restarting...

Root Cause:

- Database connectivity issues (timeouts, unreachable host) are a primary trigger for cascading failures.
- Authentication failures and invalid tokens suggest expired credentials or possible brute-force attempts.
- Unhandled exceptions (NullPointerException) in PaymentService indicate code-level bugs.
- ECONNRESET errors in OrderService point to network instability or backend failures.

Anomalies:
- Error spikes: Multiple errors within seconds (e.g., repeated database timeouts, ECONNRESET).
- Repeated failures: Authentication failed for user admin multiple times, followed by IP block.
- Resource issues: Slow requests and high latency (>5000ms) detected in API and order processing.
- Security concerns: Multiple invalid login attempts and token expiration.

Performance Findings:
- Slow requests: /api/user (1892ms), /orders (5200ms, 6300ms), general requests (8290ms).
- Latency patterns: Consistent delays in processing and completion of requests.
- Bottlenecks: Endpoints /api/user and /orders repeatedly slow.

Security Concerns:
- Authentication failures and repeated invalid login attempts.
- IP blocked due to brute-force detection.
- Token expiration and invalid token usage.

Recommendations:
1. Investigate and resolve database connectivity issues (network, firewall, DB server health).
2. Review authentication flows; implement stricter token lifecycle management and monitoring.
3. Address unhandled exceptions in PaymentService; add null checks and error handling.
4. Monitor and optimize endpoints with consistent latency/bottleneck issues.
5. Enhance security monitoring for repeated login failures; consider CAPTCHA or rate-limiting.
6. Review network infrastructure for ECONNRESET errors in OrderService.

Severity: Critical

Timeline:
- 2025-01-28: PaymentService exception and restart.
- 2025-02-09: Authentication failures, IP block.
- 2025-02-10: Database timeouts, invalid tokens, slow requests.
- 2025-02-11: Database timeouts, OrderService connection resets, slow API responses.

All structured fields are provided above for actionable troubleshooting and resolution.